PrimusV3 External Art-Net Integration Guide

This document explains how PrimusV3.6 receiver firmware expects Art-Net LED data. It is written for

third-party

If devices respond to Art-Net but colors look wrong, scrambled, or shifted, the cause is almost

always a r

Quick Reference — Workshop Hardware

In workshop kits, outputs are often labeled:

Workshop

Collar long_strip (Long Strip) 72 216 Linear strip

Workshop setups preset the collar output to 72 pixels (long_strip) via Primus Central's output

configurat

Other output types exist (short_strip, grid, extra_long_strip) but badge and 72-LED collar are the

standard v

How the Firmware Handles Art-Net

Transport

Item Value

What the firmware does with pixel data

1. Receives an ArtDmx packet.

2. Matche

There is no firmware-side brightness channel, no HTP/LTP merge logic, and no grid remapping.

Whatever

Brightness

Show dimming is done by scaling RGB values in the sender. The receiver hardware brightness is fixed

at 255. To

Default Port and Universe Layout

Each board profile assigns default output types and universes at boot. Verify on site — output types

can be ch

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v2 (ESP32 Feather V2)	Badge (8×4 grid)	0	Collar (72 px) *	1
v31 (S3 TFT + NeoPXL8)	Short strip (30 px)	0	Long strip (72 px)	1

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* V2 firmware defaults to short_strip on A1 at first flash, but workshop Primus Central presets A1

to long_st

Workshop kits typically use badge on universe 0, 72-LED collar on universe 1.

How to read the live configuration

Send ArtPoll to the node’s IP (or broadcast). The ArtPollReply Node Report contains a capability

tag:

PV3CAP1|port:type_id:universe|B:profile|IP:D|F:RIOH

Example for a workshop V1/V2 node with badge + 72-LED collar:

#0001 [0000] OK|PV3CAP1|0:4:0|1:2:1|B:v1|IP:D|F:RIOH

Parse each port:type_id:universe segment:

Segment Meaning

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Type IDs:

ID Type Pixels DMX channels (RGB)

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The Long Name field also lists active outputs, e.g. A0:Grid 8x4 A1:Long Strip.

ArtDmx Packet Format

Offset Field Notes

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14–15 Universe Little-endian universe number

16–17 L

Channel numbering

The first data byte (offset 18) is DMX channel 1 in Art-Net terms.

For a 72-pixel collar:

LED index DMX channels Bytes in packet

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0 1–3 R, G, B

1 4–6 R, G, B

… …

71 214–216 R, G, B

For a 32-pixel badge:

LED index DMX channels Bytes in packet

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0 1–3 R, G, B

… …

31 94–96 R, G, B

All workshop outputs fit in one universe (well under the 512-channel limit).

Color order in the packet

Send RGB — red first, then green, then blue — for every pixel.

The firmware calls the NeoPixel library with (R, G, B) and uses NEO_GRB on the wire. You do not need

to pre-swa

Collar (Long Strip) — Linear Pixel Order

The workshop collar is a 72-LED linear strip (long_strip). Pixel index follows the physical wiring

from the d

Data in ──► [0] [1] [2] ... [70] [71]

Art-Net mapping is straightforward: channel 1 = LED 0 red, channel 2 = LED 0 green, channel 3 = LED

0 blue, ch

EOS collar checklist

- Fixture mode: RGB (3 channels per pixel), not RGBW, not CMY.

- Pixel cou

Badge (Small Grid 8×4) — Serpentine Pixel Order

The badge is 32 pixels arranged as an 8 column × 4 row grid, but the LED strip is wired in

serpentine

- Even rows (0, 2): left → right

- Odd rows (1, 3): right → left

Physical LED index map (numbers are pixel indices in the Art-Net stream):

Row 0:   0   1   2   3   4   5   6   7

Row 1:   1

Row 0 is the row where the strip starts (data-in). Confirm orientation on the physical badge if

left/right s

What Primus Central does (and EOS must match)

Primus Central designs effects in logical row-major order (row 0 left→right, row 1 left→right,

etc.), then

If EOS is patched as a simple 32-pixel line, or as an 8×4 matrix with every row left→right, even

rows may

EOS badge checklist

- Total pixels: 32 (96 RGB channels).

- Layout: s

Why Colors Look Unpredictable — Common Causes

All pixels show a shifted rainbow or wrong solid   Wrong start channel or extra channel before pixel

1   Sta

Pattern appears on wrong device   Wrong universe or wrong destination IP

Read PV3

Collar looks correct, badge rows zig-zag   Badge sent as linear or progressive grid, not

serpentine

Only first few LEDs change   Pixel count too low in patch (e.g. 30 instead of

72)   Se

Collar pattern repeats or wraps oddly   Collar patched as 30 px (short_strip) instead of

72   Co

Colors flicker or stutter   Multiple senders fighting, or sequence issues

One contr

Everything very dim   RGB scaled low; no separate brightness channel

exists   I

Collar responds on badge universe Output types changed from defaults

Re-read A

EOS-Specific Setup Notes

EOS terminology varies by version and network configuration. Map these concepts to Primus:

Universe 0, 1	Net universe (confirm whether your net uses 0- or 1-based

display)

32 or 72 RGB pixels	LED fixture, pixel map, or RGB array
96 or 216 channels	Total DMX footprint per universe output
Serpentine 8×4	Matrix wiring / snake order / pixel map

Recommended verification procedure

Run these tests before building cues:

1. Discovery — ArtPoll the node; confirm PV3CAP1 segments and note each port's universe and type

ID.

2. Collar solid red — Universe for A1 (usually 1). Send LED 0 = (255, 0, 0), all others off. The

first LED a

3. Collar walk — Light one pixel at a time in index order 0→71. Confirm sequential motion along the

strip.

4. Badge corner test — Universe for A0 (usually 0). Light only pixel 0 green. Only one corner LED

(start of st

5. Badge serpentine test — Light pixel 7 blue and pixel 8 red simultaneously. They should be

adjacent a

6. Badge row reversal test — Light pixels 8–15 one at a time. They should run right to left along

row 1 (ind

Network

- Primus nodes join WiFi (default SSID in firmware is often OPERADEV unless reflashed).

- EOS and

Frame Timing

- The node renders as packets arrive; target ~30 FPS for smooth motion.

- When bo

Protocol the Firmware Does Not Use

Sending these will not fix color issues:

- No sACN (E1.31) — Art-Net only

- No separ

Primus-specific opcodes (0x8100 output config, 0x8200 IP config, 0x6000 rename) are for management

only; they

Related Documentation

- Full protocol reference: API_REFERENCE.md (ArtDmx, output types)

- Firmware

Summary

Output

Collar	2	72	216	1 (A1)	Linear 0→71	RGB

The firmware is a thin transport layer: universe → byte copy → LED index. Predictable color from EOS

requires m